

Theory Assignment 2

24M0319, 24M0326

$$1. \quad \mathcal{L} = \sum_{n=1}^N \|x^{(n)} - \hat{x}^{(n)}\|^2 \quad \begin{matrix} W \in \mathbb{R}^{k \times d} \\ V \in \mathbb{R}^{d \times k} \end{matrix}$$

$$\hat{x}^{(n)} = V(Wx^{(n)} + b) + c$$

After removing the ^{mean} offset

$$\hat{x}^{(n)} = VWx^{(n)}$$

$$\mathcal{L} = \sum_{n=1}^N \|x^{(n)} - VWx^{(n)}\|^2$$

Let's fix any V, W and consider the mapping $x \mapsto VWx$.

The range of this mapping, $R = \{VWx : x \in \mathbb{R}^d\}$, is a k dimensional linear subspace of $\mathbb{R}^d$.

Let $U \in \mathbb{R}^{d \times k}$ be a matrix whose columns form an orthonormal basis of this subspace, i.e. range of U is

R and $UTU = I_{k \times k}$. Therefore each vector in R can be

written as Uy where $y \in \mathbb{R}^k$. For every $x \in \mathbb{R}^d$ and $y \in \mathbb{R}^k$

we have

$$\|x - Uy\|_2^2 = \|x\|^2 + y^T U^T U y - 2y^T U^T x$$

$$= \|x\|^2 + \|y\|^2 - 2y^T U^T x \quad | \because U^T U = I_k$$

So minimizing $\|x - Uy\|^2$ w.r.t. y gives us

$$\frac{\partial}{\partial y} \|x - Uy\|^2 = 0$$

$$\Rightarrow 2y - 2U^T x = 0$$

$$\Rightarrow \boxed{y = U^T x}$$

Therefore for each x we have

$$\|x - UU^T x\|_2^2 = \min_{\tilde{x} \in R} \|x - \tilde{x}\|_2^2$$

So, we can replace V, W by U, U^T for optimal solution.

and so conclude

$$\boxed{V = W^T} \text{ is the optimal soln}$$

$$\text{and } V^T V = I_{k \times k} \quad \text{or} \quad W W^T = I_{k \times k}$$

columns are orthonormal, rows are orthonormal,
orthonormal

2.

From ① we saw that the optimal soln is

$$V^* = (W^*)^T$$

and $W^* (W^*)^T = I_{k \times k}$ (rows are orthonormal)

$$\text{Now } L = \sum_{n=1}^N \|x^{(n)} - W^T W x^{(n)}\|^2$$

$$= \sum_{n=1}^N \|x^{(n)}\|^2 - 2 x^{(n)T} W^T W x^{(n)} + x^{(n)T} W^T (W W^T) W x^{(n)}$$

$$= \underbrace{\sum_{n=1}^N \|x^{(n)}\|^2}_{\text{fixed}} - \sum_{n=1}^N x^{(n)T} W^T W x^{(n)} \quad | \quad W W^T = I$$

So we

$$\min L \equiv \max \sum_{n=1}^N x^{(n)T} W^T W x^{(n)}$$

$$= \max \text{Tr} (X^T W^T W X)$$

$$= \max \text{Tr} (W (X X^T) W^T)$$

Now $W (X X^T) W^T$ is eigen decomposition of $X X^T$ and since all K rows of W are orthonormal, the trace maximization of the trace means picking the largest K eigenvalues, which is equivalent to PCA.